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Chapter 20
Fostering Research into Coupled Long-Term Dynamics of Climate, Land Use, Ecosystems and Ecosystem Services in the Central French Alps

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Abstract The Central French Alps long-term socio-ecological research Platform (Central French Alps LTSER) focuses on the coupled dynamics of alpine ecosystems, their uses and climate. The creation of the Platform has provided a unique opportunity to initiate and strengthen collaborative transdisciplinary research involving a range of natural and social scientists (ecologists, agronomists, climatologists, sociologists) and key regional stakeholders from the agriculture, tourism and nature conservation sectors. The main research questions were built on existing long-term research projects at two sites. They include climate change effects on biodiversity and ecosystem functioning, and coupled dynamics of grassland management, biodiversity and ecosystem functioning through ecosystem services, using not only observations of

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natural and human systems, but also manipulative experiments of climate, management and plant and soil diversity to feed models. The LTSER Platform has fostered three important types of advances: (1) Long-term data consolidation and sharing. (2) Invigorating interdisciplinary projects (e.g. coupled transformations of economic functioning of farming systems and mountain summer pastures dynamics; mutations of alpine tourism in the face of climate change). (3) New transdisciplinary projects, including climate change adaptation of mountain territories, integrated carbon cycle modelling in response to historical land-use change and climate; a sociological study of the process of construction of the LTSER Platform.

Keywords Mountain system • Land use dynamics • Climate change • Biodiversity • Ecosystem services

20.1 Introduction

In the last few decades, social and ecological scientists have worked and developed their theories in more or less complete isolation from each other. While the former have often thought "as if nature did not matter" (Murphy 1995), the latter were particularly interested in the biological and physical dimension of nature. Moreover, human influence was and is still today often interpreted by ecologists as a "perturbation" (White and Jentsch 2001) and is more rarely seen as part of the complex interactions in nature (Liu et al. 2007), while many sociologists regard nature as a purely social construct. Today's environmental challenges, in particular the effects of global change and the predicted considerable human impact via land-use changes on biodiversity (Sala et al. 2000) require in particular that sociologists, economists, climatologists and ecologists develop common approaches in order to collaborate on one of the most pressing problems of the twenty-first century. Recognition has grown in all fields over recent years that neither human nor biophysical systems can be studied autonomously. This recognition is the impetus for studying the dynamics of complex and interdependent social and ecological systems as it is undertaken in LTSER research (Haberl et al. 2006).

In France, several research institutes and universities decided to participate in the creation of an LTSER Platform in the Alps in order to address the complex question of interactions between humans and the environment, in the context of mountain systems. Here, we describe and reflect on the process of creation and implementation of a newly established long-term socio-ecological research Platform in the French Alps seeking to promote the integration of interdisciplinarity in research that initially focused on ecology. The project emerged simultaneously with a Europe-wide initiative to create and implement LTER sites and LTSER Platforms (Mirtl 2010), but the construction process took place in parallel, rather than being driven by the European initiative.

We first describe the process of creating the LTSER Platform, including how the genesis of the LTSER Platform itself became an object of research for sociologists,

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and the key collaborative tools that were built to facilitate data exchange. We then illustrate with recent research projects how the creation of a long-term socio-ecological research Platform provides an important framework to formalise transdisciplinary collaboration.

20.2 Constructing the Central French Alps LTSER Platform

20.2.1 Identification of Potential Partners

When the idea of creating a LTSER Platform first emerged, both lead partners, CEMAGREF (Institute of agricultural and environmental engineering research; a government-funded research organisation under the administrative authority of France's Ministries of Agriculture and Research) and CNRS (a government-funded research organization under the administrative authority of France's Ministry of Research), were already involved within their respective organisations in establishing their own priority research area in the French Alps: CEMAGREF within the framework of Alter-Net, while CNRS was integrated in the French scheme of the "Zones Ateliers" (literal translation: "workshop areas"). In each of these organizations, one laboratory was particularly involved: the Alpine Ecology Laboratory (LECA) at CNRS, and the Mountain Ecosystems Research Unit (RU EM) at CEMAGREF. As they were already collaborating as part of ongoing projects and regional infrastructure development, the idea of merging the two processes into a larger one was rapidly adopted. However, as both laboratories within each institute were largely dominated by ecologists, significant effort was made to enlist researchers from other disciplines in the project (Table 20.1). In a second step, non-scientific actors such as local representatives from forestry and agriculture, and important stakeholders in the field of nature conservation including national and regional parks were contacted (Table 20.1).

20.2.2 Choice of the Main Research Topic and the Study Region

20.2.2.1 The Study Region

The Central French Alps have a number of key assets for the study of the dynamics of land systems in the context of global change. It is well known that mountain regions in general, and the Alps in particular, have developed under strong natural constraints. The rugged topography as well as the north–south direction of the ranges results in steep environmental gradients, as well as in a high spatial heterogeneity in climatic, topographic and soil conditions. Such constraints promote the existence of a large diversity of natural habitats, making the Alps a biodiversity hot

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Table 20.1 Partners and their disciplines involved in the Central French Alps LTSER Platform Aps

Scientific partners	Disciplines
CNRS Alpine Ecology Laboratory*	Ecology
Cemagref Research Unit (RU) Mountain Ecosystems*	Ecology, agronomy, forestry
CNRS Transfers, Hydrology and Environment Laboratory	Climatology, hydrology
CNRS Alpine Research Station Joseph Fourier	Ecology
Cemagref RU Development of Mountain Areas	Agronomy, sociology, economics
Cemagref RU Public policies, politic actions and land management	Geography
Local and socioeconomic partners	
Ecrins National Park	Nature conservation, education
Vercors Regional Nature Park	Nature conservation, education
Nature Reserve Hauts Plateaux du Vercors	Nature conservation
National Alpine Botanical Conservatory	Nature conservation, botany
Several professional bodies in the field of agriculture	Agriculture
Office for Agriculture of Hautes-Alpes	Agriculture
National Forest Bureau	Forestry

Lead partners are indicated with an asterisk

Box 20.1 Structure and Governance of the Central French Alps LTSER Platform

Zone Atelier Alpes, the Central French Alps LTSER Platform, was established jointly by CNRS and CEMAGREF in 2009, following a 2-year construction process based on exchanges between both these two institutions and the leading laboratories, and a participative construction process among academic partners, and between these partners and key regional stakeholders (see Table 20.1 for the list of academic and non-academic partners). French Zones Ateliers are an administrative entity of CNRS, which are certified for a period of 4 years, with an annual review and a full re-examination, leading to continuation, restructure or even discontinuation, every 4 years. As such they receive an annual budget (ca. 50 k€/year in the case of Zone Atelier Alpes, based on pooled contributions from CNRS and CEMAGREF), allocated on a year-to-year basis depending on a requested budget and availability of funds, and which they then manage in a subsidiary manner.

In the case of the Central French Alps LTSER Platform, the scientific strategy and the budget are managed by a steering committee made of the two coordinators (currently, one from CNRS and one from CEMAGREF) and the three scientists in charge of each of the scientific axes (see main text – Sect. 20.2.2.2, "Scientific Questions"). This group communicates on a regular basis by email for the management of projects, information dissemination and other issues such as data management, and meets on a regular basis (about

(continued)

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Box 20.1 (continued)

every trimester) to monitor the implementation of scientific strategy, organise internal meetings and meetings with outside scientific or societal parties, and coordinate the budget and its allocation across projects and other actions. The management structure was designed to be light and flexible so as to adapt to the day-to-day dynamics of research. In addition, a formal meeting with representatives of each of the Platform partners (Table 20.1), including in particular non-academic partners, is organised yearly. Annual science meetings, when possible associated with the field trips, are organised to enable sharing of project results. In addition, dissemination meetings for regional stakeholders from the different sectors relevant to the project (agriculture, forestry, nature conservation, tourism) have taken place on a 2-yearly basis.

The Platform and the projects that are run therein involves a total of 11 full-time equivalent (FTE) scientists, 6 FTE technical staff and 17 FTE contract staff (mostly Ph.D.s and postdoctoral students but also some short-term technical staff). It draws upon an annual operating budget for research projects (salaries of permanent staff not included) which amounts on average to 1.5 million EUR per year, and comprises funds obtained from the local (e.g. Conseil Général de l'Isère, Région Rhône-Alpes), national and European levels.

spot (Körner 1999). This is particularly true for the Central French Alps, given their geographic position at the boundary between the external Alps with rather more Atlantic conditions, continental influences in the internal Alps, and those areas with Mediterranean influences.

At the same time, these natural constraints have been reflected in human development, and especially in agricultural development, in, for example, obstacles to mechanisation and limited competitivity in trade due to high labour requirements and transportation costs. Over history, climatic and other natural constraints and their spatial heterogeneity have become integrated into cultural diversity and activities such as agriculture, forestry, tourism, and hydroelectricity (Tappeiner and Bayfield 2004). Therefore, we hypothesise that ecological and human co-adaptation has taken place. Among other features of the coupled human-environment system, important aspects that have shaped today's landscapes include: (i) climate variability over a large variety of scales (centuries to interannual); (ii) the development of land-use systems that are tightly linked with the provision of a variety of ecosystem services (Girel et al. 2010); and (iii) a strong multi-functionality and regional diversity in land-use systems, but also competition among alternative uses. Such complex interactions among climate, ecosystems and land-use systems raise the question of whether the Central French Alps land system is expected to be resilient or vulnerable to global environmental change, and which adaptation strategies may be developed to respond to environmental as well as to socio-political changes.

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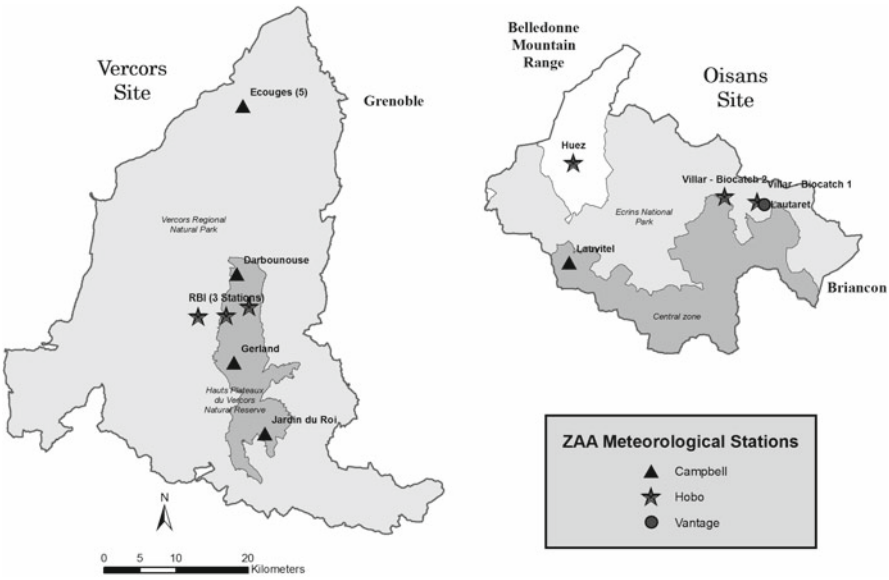


Fig. 20.1 Location map for the Central French Alps LTSER Platform and meteorological stations. The Platform includes areas with strict protection status (*dark grey*): the Vercors High Plateaux Natural Reserve and some of the core area of the Ecrins National Park, as well as inhabited areas managed by agriculture and forestry (*light grey*): the Vercors Natural Regional Park and part of the peripheral area of the Ecrins National Park. Meteorological monitoring stations set up by the LTSER Platform are located using different symbols depending on the equipment in place

The specific study areas for the Central French Alps LTSER Platform were selected among sites already studied by at least one of the institutional partners, and thus where significant data sets were available and much knowledge had already been acquired regarding ecological (for the main part), climatic, economic, and social processes. We cross-checked the most promising regions with the recommendations given by Ohl et al. (2007). Finally, two study regions with three main monitoring sites were selected, and form the Central French Alps LTSER Platform (Fig. 20.1). In economic terms as well, the Platform contains a wide diversity. Intensification and extensification processes are ongoing within the agricultural sector, and, with the ski resort "Alpe d'Huez" and the Ecrins National Park, the tourist sector is highly developed. Moreover, site-specific policies addressing biodiversity issues are implemented and governmental investments in biodiversity conservation are present as a national park (Ecrins) and a regional park (Vercors) which are included in the Platform. Finally, the Platform encompasses two NUTS 2 levels (Rhône-Alpes and Provence-Alpes-Côte d'Azur) and three NUTS 3 levels (departments of Isère, Drôme and Hautes-Alpes) which may help us to understand how European or national decisions may influence ecological and socioeconomic factors at a regional level.

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20.2.2.2 Scientific Questions

It seems important to emphasise that the choice of the core research theme and of the study areas was a dynamic process, in which several other partners were contacted and the research topic and the study area redrawn several times. However, we only describe the process and its outcome, and deliberately leave out (in spite of the substantial efforts entailed) any description of the processes required to obtain funding and resources.

Right from the beginning, the intention of all partners was to create not an observatory of current environmental and social development, but an area where most of the research is and will be carried out. We therefore deliberately favoured a bottom-up approach, where the core research topic was developed based on the interests of the majority of the participants and on the characteristics of the study area, and not, as proposed by others, imposed following a more academic, top-down, approach (Redman et al. 2004). All partners agreed that research themes may not be permanent, but will evolve as more data and knowledge are accumulated concerning the study system. The core theme finally selected is "research on coupled dynamics of alpine ecosystems, their uses and the climate". It has already consolidated over the 3 years since the Platform was established, with transdisciplinary projects converging towards the analysis of processes of adaptation of alpine territories to global change and societal transformations (Fig. 20.2). Three main research axes were then jointly selected to implement this core theme, and consider (i) the assessment of climate changes and its impacts on biodiversity and uses, (ii) the analysis of the coupled dynamics of grassland management, biodiversity, and ecosystem services, and (iii) the development of new interdisciplinary studies in the study area. This joint approach is implemented with a simple governance structure where each research axis is coordinated by a climatologist, an ecologist, and a sociologist respectively.

20.2.3 Data Pooling and Standardisation as a Key Element of Success

20.2.3.1 Climate Monitoring and Climate Data

The quantification of spatial and temporal climate variation is essential to the characterisation of the studied environments, both in terms of quantifying climate trends and of providing essential variables for the analyses of biodiversity, biophysical processes and resources for human use. Since 2006, the Central French Alps LTSER Platform has developed a network of stations to monitor climate at the three intensive research sites and to characterise key gradients of variation (Fig. 20.1).

Such data will make it possible to produce fine-scale decadal analyses of climate trends, as was undertaken at the regional scale for the Vercors central plateau (Fig. 20.3). There the analysis of long-term data series from Météo-France stations

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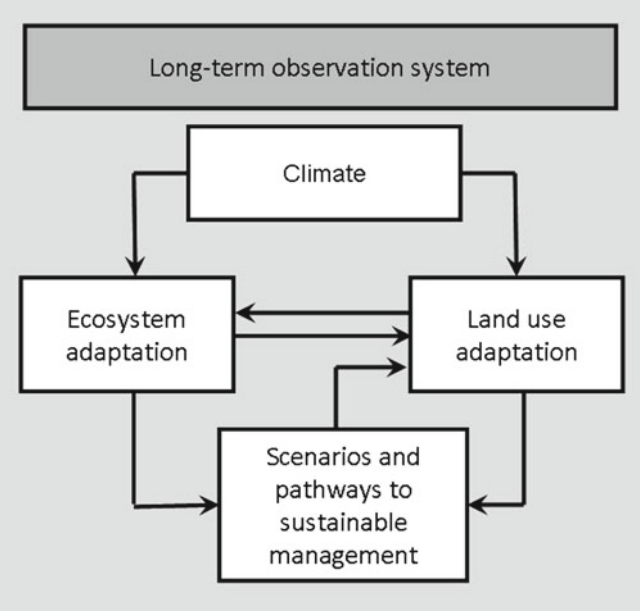


Fig. 20.2 Conceptual presentation of the Central French Alps LTSER research questions

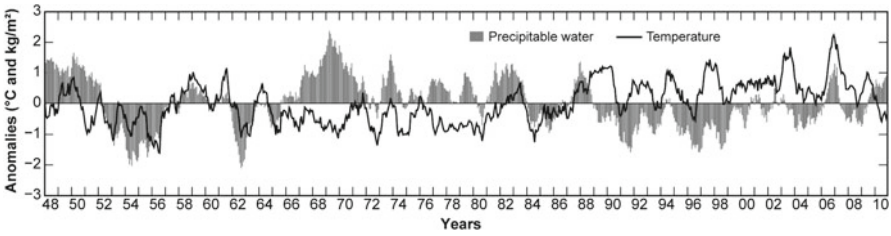


Fig. 20.3 Recent climatic trends over the Vercors High Plateaux. Interannual variability of air temperature (at 850 hPa level; in °C) and water precipitation (in kg/m^2) calculated for the Vercors site on the 1948–2010 period (anomalies were calculated from NCAR-NCEP reanalysis; time-series anomalies are smoothed using a moving average over 12 months)

within the study region demonstrated a clear warming trend of +1 °C over the last 30 years, while there was no clear trend for total annual precipitation in spite of a shift to more Mediterranean regimes (Bigot and Rome 2010). The new network will complement such analyses with stations at high altitude, making it possible to incorporate the effects of complex terrain into regional-scale climate analyses, as well as providing detailed information to be linked with other social and ecological variables monitored at the study sites.

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20.2.3.2 Plant Species Distributions and Plant Functional Traits

The understanding of the spatial distribution of plant species and their functional attributes in relation to environmental drivers requires a comprehensive distribution and plant functional trait database. The Alpine National Botanical Conservatory (CBNA) and the Alpine Ecology Laboratory (LECA) spent a considerable amount of time crosschecking and validating existing vegetation surveys from the original CBNA database, in order to keep only recent (>1980) and sufficiently precise (resolution >200 m) survey plots. LECA maintains this subset of the CBNA database with a comprehensive taxonomy (and associated synonymy), knowledge of the local habitat of each survey plot and percentage of bare-soil. In the mean time, LECA has coordinated the development of the ANDROCASE plant functional trait database that aims to collect existing plant functional traits from various sources (open database, literature, own measurements) for the regional flora of the entire French Alps (3,400 taxa). CBNA but also the Ecrins National Park have participated in developing and contributing to the database. ANDROSACE now contains about one million entries and some traits (e.g. maximum vegetation height, reproduction type, life form) or ecological indicators (Ellenberg values) have been collected for all known species in the region. The area covered by the Central French Alps LTSER Platform encompasses 1,457 plant species, present in a total of 4,717 georeferenced botanical surveys. For these species, trait values are known exhaustively for common traits such as plant height, phenological types, as well as their ecological preferences (Ellenberg or Landoldt indices). Traits requiring more intensive data collection are for instance SLA which is known for 687 species, LNC for 207 species, or seed mass for 999 species, using both data collected from Platform sites and sourced from external data bases.

20.2.3.3 Coupling Botanical and Agronomic Data on Alpine Grasslands to Investigate Land-Use and Climate Change Responses

Observed vegetation changes between two sampling periods may reflect responses to changes in land-use, but also to climate. The Research Unit Mountain Ecosystems (EMGR) started in the 1970s a systematic inventory of summer grazing pastures in the Vercors in order to characterise the vegetation and its trajectories. Relevés were repeated more or less regularly and have recently been compiled into the database FlorEM containing more than 5,000 geolocalized line transects and plot data with records of the abundance of each species. In addition, at each of the two sites of the Central French Alps LTSER 12 paired permanent plots, one grazed and one where grazing has been excluded for more than 20 years (Haut-Plateaux du Vercors) or 30 years (Alpe d'Huez), have been monitored over the years. These plots may serve as a baseline under conditions where the vegetation follows its natural successional trajectory only influenced by current climatic conditions and past grazing patterns.

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However, as little expertise for assessing agro-economic aspects was available at RU EMGR, changes in land use were not always recorded. Recently, agronomists from the RU Development of Mountain Areas have repeated field investigations already conducted several years ago in order to assess changes in the management and use of summer pastures and of lowland pastures used by the same herds. These data are also stored in a spatially explicit database.

Further research will link these two currently independent databases in order to provide a ready overview of changes in the vegetation and, in areas where socioeconomic data is available, information about changes in land use of the summer pastures under study. In a further step, these two databases should be linked to the already existing climatic database in order to distinguish land use from climate signals in the observed vegetation dynamics.

20.2.3.4 Building a Meta-Data Base

In order to facilitate the comparison of data sets on different components of the human-environmental system, one of the first common actions in the LTSER Platform was the creation of a meta-database (MDB). The description of existing data within the meta-database provides a first approach to supporting and promoting collective use of the data collected by different partners. In particular, the quality of old data we wished to analyse in a new context (global change) was quite heterogeneous, especially because methods have evolved and original data sets were collected in contexts that have changed. In the future, data collection will be planned and coordinated as much as possible by standardising measurements, collectively establishing protocols, and sharing data management tools. The MDB also aims to rationalise the researchers' activities through avoiding the duplication of work or the search of data from external sources, without knowing it has already been conducted by somebody else. For example, current efforts focus on inventorying and geo-localising botanical and agronomic data within a common GIS Platform, and on consolidating a database of remote sensing and aerial photographic images.

20.2.4 Integrated Approaches from Field Observations to Experiments and Modelling

One of the strengths of the French Central Alps LTSER Platform lies in the integration of the full range of approaches from field observations and monitoring to experiments and modelling. The Platform includes monitoring of climate (see above), land use, vegetation and a range of ecosystem processes such as biomass production, soil water transfers and fertility over a set of permanent grasslands plots located across the three sites. This data baseline is complemented by spot surveys, e.g. vegetation and soil maps, over entire sites, which we aim to repeat on a 30-year basis. More intensive experimental field manipulations address the effects of specific drivers such as grassland management (through the use of long-term

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grazing exclosures – e.g. two experiments running for over 30 years), mesotopography (comparisons of ridges and snowbeds), or climate change (climate warming and drought experiments) on vegetation composition and biogeochemistry (C and N cycles), as well as on agronomic characteristics such as digestibility or phenology of grass flowering. These manipulations are complemented by garden experiments, which aim to identify detailed ecophysiological, demographic or biochemical mechanisms and are made possible by the Lautaret Alpine Botanical Garden located at 2,100 m, which allows experiments under realistic mountain conditions. Finally long-term data analyses and process studies feed into modelling, including macroecological modelling, ecophysiological modelling and spatial modelling of ecosystem services (Baptist and Choler 2008; Albert et al. 2010; Lavorel et al. 2011).

20.3 The LTSER Platform Itself Becomes a Research Object

20.3.1 Objectives and Research Questions

The establishment of the LTSER Platform has been accompanied by a reflexive evaluation of the process whereby the LTSER Platform itself is seen as a research object. The LTSER Platform constitutes a very good case in which to study scientific cooperation in practice, from a science and technology studies (STS) perspective, looking at "science in the making" rather than in "ready-made science" (Latour 1987; Knorr-Cetina 1999). Indeed, several types of scientific cooperation are present: cooperation between research groups tackling closely related research questions in ecology but belonging to organisations focusing on 'fundamental' (CNRS) or 'applied' (CEMAGREF) research; cooperation between various disciplines, within natural sciences but also between natural and social sciences; cooperation between scientists and nature managers; cooperation among scientists themselves, with different research projects and individual aims. Beyond the claim that cooperation is good for science, we aim at understanding how, to what extent, and under which conditions it takes place, and at understanding its effects. We addressed specifically the question of whether the LTSER Platform brings something new, in the context of an overall objective of our current work of analysing how the climate and biodiversity 'crises' transform ecological research.

20.3.2 Methods

We used a survey relying mainly on semi-structured interviews with the members of the LTSER Platform, during which the following topics were addressed: history and assigned objectives of the structure, biographical information on the interviewee,

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creation of the Platform and personal involvement, ongoing activities with the Platform, expectations and difficulties encountered. A targeted survey focused on the relationships between the CBNA and CNRS scientists around the exchange of botanical data and the construction of the ANDROSACE data base (see above). We also drew on various other materials such as participating as observers during meetings, analysis of communication and of documents in order to analyse the perspectives of different partners and individual participants, the topics of debate, and meeting and interviewing participants' about their expectations. For more details on the interviews, see Granjou and Mauz (2011) and Mauz and Granjou (2011). Here we briefly summarise outcomes in terms of reported perceptions on how the LTSER Platform started, why it started and what it changes in the type of work and working relationships among scientific partners, and with partner nature managers.

20.3.3 Key Lessons

How it started: The interviews revealed that a meeting was organized by the Ecrins national park as early as 2004, hence long before the project of creating a LTSER Platform was considered. This meeting gathered researchers from CNRS and CEMAGREF and staff of the park's scientific service, hence anticipating the association of different groups of actors (researchers, nature managers and some local stakeholders) in the Platform (Table 20.1). Citing a lack of cooperation between the research teams in ecology working on its grounds, the park was eager to promote exchanges and offered itself as a possible common research field. However, one of the labs had just been restructured and felt a need to consolidate its own research project before embarking on a broad cooperation with the other labs. It therefore took some time before building the Platform as a medium-term, cooperatively constructed process. Similarly, the Vercors reserve managers expressed the same need for scientific work on its territory to be promoted and rationalised, which illustrates the importance of the nature managers' initiative in this process.

Why it started: A majority of ecologists involved in the LTSER Platform seek to understand ecosystems' responses to changes in climate and land use. They rely very much on long-term data and access to databases, unanimously described as difficult and costly, which is thus undoubtedly one of the main aims of the Platform. For that matter, the construction of a meta-database has been one of the first activities carried out by the Platform (see Sect. 20.2.3). However, the partners' relationship to data is rather ambivalent: their need for data encourages them to focus on a few common research sites, preferably where long-term[1] data are available, and to

[1] It is important to note that "long-term" does not mean the same for all participants. Ecologists working on plant adaptation to climate change and land-use changes may need data extending back to the nineteenth century, whereas a single decade may be considered as "long-term" by researchers studying phenomena developing over short spans of time.

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pool and share their data with their colleagues. On the other hand, they are sometimes reluctant to hand over data obtained through a long period of painstaking work, which, as Bowker states, is certainly not "raw" (Bowker 2000, p. 670). The meta-database illustrates this ambiguity: it records what kind of data is available without giving access to the data itself, and the conditions under which these can be exchanged, shared and used are negotiated between individuals. Nevertheless the construction of the relationship, e.g. between CBNA and CNRS, illustrates the process of mutual building of trust. It is hoped that this example will pave the way for facilitated data exchange and access between members of the LTSER.

What it changes: For natural scientists, the LTSER Platform seems to affirm existing relationships and data exchanges more than it creates new ones. The collaborations were already active between at least some of the natural scientists, and between them and the nature managers. For example, young researchers from the two main labs have worked together since their PhD studies, and an agreement was signed by the LECA and CBNA in 2007, by which the lab has obtained access to the conservatory's exceptional floristic database, under agreed conditions. According to many participants, collaborations were hence already well under way and the Platform construction does not appear to have been a revolutionary event. However, official labels, such as a "Zone Atelier"[2] and a LTSER Platform, are said by some respondents to be important in terms of visibility and recognition. This sign of recognition is particularly important for nature managers, who thereby obtain official acknowledgement of their direct involvement in research projects.

The situation is different with social scientists, since there was very little previous collaboration: the Platform could theoretically play an important role for them. However, the recruitment of social scientists into projects has turned out to be problematic, despite the efforts of the Platform leaders and despite the existence of the aforementioned projects. Moreover, interdisciplinarity between social and natural sciences seems scarcely desired by many. Significantly, an ecologist describing this relationship used the term "forced marriage": it is seen as imposed from the top rather than pushed from the ground up and researchers underline that this kind of interdisciplinarity is difficult to achieve, demands very firm disciplinary grounds and does not lead to easily publishable results. Some interviewees recollected unpleasant experiences, where they had submitted research projects with a clear interdisciplinary orientation that had been rejected (contrary to more monodisciplinary projects), although the calls had strongly encouraged interdisciplinary contributions. Interdisciplinarity was therefore regarded as a rather risky adventure upon which only well-established scholars could afford to embark.

[2] Zone Atelier sites make up the French LTER network, coordinated by CNRS's Environment and Sustainable Development Institute. Currently there are ten sites, whose membership of the network and management are the responsibility of an international Scientific Committee. The Zone Atelier network promotes long-term interdisciplinary research at sites representative of the main French river basins and of different type of natural and managed environments. See: http://www.za.univ-nantes.fr

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20.4 Emerging Transdisciplinary Projects of the Central French Alps LTSER Platform

In spite of these initial teething pains, the process of collectively constructing inter- and trandisciplinarity through the LTSER Platform has been fruitful. New projects offer an added-value to the sets of already ongoing disciplinary projects, which they are expected to reinforce in the long term. The construction of the Platform has made it possible to retrieve "old" collaborative projects sometimes designed for other study areas, which will now use the LTSER Platform as a study area. Other projects have been created *ex nihilo* thanks to the existence of the Platform. In the following, we briefly describe three projects which have already been implemented.

20.4.1 Integrated Analyses of Ecosystem Services

Ecosystem services have recently developed as a concept that makes it possible to interface the ecological analysis of natural or managed systems and analyses of their uses and associated human dynamics (Turner et al. 2007). Recent reviews and individual studies have highlighted the complexity of ecosystem services analyses in that they require the availability and coupling of a large variety of types of data including land-use/land-management data, abiotic data e.g. on topography and soils, biodiversity data and data on human uses and preferences (de Groot et al. 2010). A LTSER Platform thus appears to be an ideal basis for detailed analyses of ecosystem service provision and its relationships to demand by stakeholders.

We quantified ecosystem services provided to farmers and other locals, as well as to tourists, by the south-facing grasslands of Villar d'Arène. The site's land-use history since the eighteenth century has been documented and is referenced in a Geographic Information System (Girel et al. 2010). Locally relevant ecosystem services were identified on the basis of interviews with local farmers on their needs from and uses of grasslands (Quétier et al. 2010). Grassland functions identified by stakeholders were translated by researchers into measurable indicators (Quétier et al. 2007). Vegetation, plant functional trait and environmental data, including soil parameters, have been collected since 2003 for 57 permanent plots stratified by land-use type, landscape sector, and altitude. We measured biomass production, biomass Crude Protein Content (CPC) and digestibility, litter accumulation, flowering phenology (date of flowering onset and duration of flowering), and then calculated indices of plant species and functional diversity. This data allowed us to build statistical models for the supply by the grasslands of the range of services of interest to the stakeholders including agronomic value, pollination, soil carbon storage and cultural value from aesthetics and conservation of plant diversity (see

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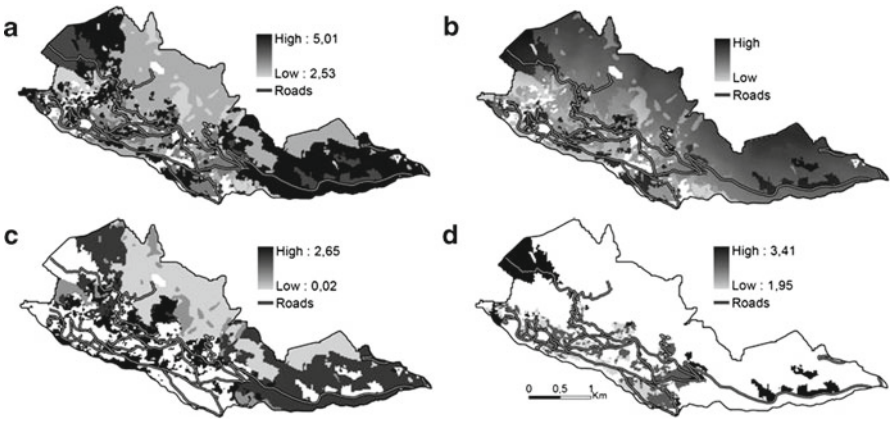


Fig. 20.4 Potential ecosystem service supply and actual provision of agronomic services to farmers of Villar d'Arène (Hautes Alpes). (**a**) Green biomass (tons/ha) (**b**) Potential agronomic value (unitless) calculated as a combination of different functions (green biomass, digestibility and phenology), and actual benefits: (**c**) the number of days of livestock units/ha and (**d**) hay production (tons/ha). Roads and tracks are added on maps as they are important elements of analysis (Modified from Lamarque et al. 2011)

Lavorel et al. (2011) for details of the field measurements and models) (Fig. 20.4). These analyses also highlighted tradeoffs and synergies among these different ecosystem services. Traditional land uses such as organic fertilisation and mowing, or high altitude summer grazing, were linked with ecosystem services hot spots because functional characteristics supporting fodder production and quality are compatible with species and functional diversity. By building on state-of-the-art ecological knowledge, our analyses made it possible to relate patterns of association among ecosystem services to the dominant plant traits underlying different ecosystem properties. For example, the functional decoupling between height and leaf traits provided alternative pathways for providing high agronomic value (through high biomass production and/or high quality and delayed flowering), as well as determining hot and cold spots of ecosystem services. Finally, we were able to compare this estimated supply of ecosystem services to the actual uses made by the farmers for grazing and hay making – i.e. to actual ecosystem service provision, showing that, depending on grassland location in the landscape with respect to the village and to altitude, and ultimately to actual use as part of the farming systems, different dimensions of modelled agronomic value such as biomass production, quality or timing of flowering determined use value to the farmer (Lamarque et al. 2011).

Similar analyses coupling historical land-use analysis, vegetation and plant trait data and estimates of the components of carbon cycling through ground-based measurements and remote sensing are in progress for the Vercors Plateau as part of a new funded project.

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20.4.2 Adaptation of Mountain Livestock Farming Systems to Socioeconomic and Climate Change

The LTSER Platform Central French Alps began with the asset of a large amount of pre-existing data from ecological monitoring, in particular for summer pastures which host flocks managed by transhumance every summer. Several studies on management practices have afforded us a good knowledge of the functioning and of the organisation of mountain summer pasture management, but little is known about the role and value of these summer pastures as part of farm system management. However, the socioeconomic environment has undergone and is still undergoing significant changes (Europe's new Common Agricultural Policy, volatility of prices for agricultural raw materials, land access difficulties, urbanisation, etc.), as well as farm organization (increasing labour costs, less manpower due to competition or complementarity with other activities within the household). In this context, one can hypothesise that farming systems will change radically, which in turn could have important consequences for the management of mountain summer pastures with resulting substantial changes in their biodiversity and the ecosystem services they provide (see below). At the same time, climate change and the associated dynamics of natural habitats may also induce changes in farm management.

This study involves several areas of expertise represented by different specialists within the LTSER Platform. Agroeconomists study the role of the mountain summer pastures as part of farm systems, and on socioeconomic changes within recent years and their impacts on entire farm systems, whether located within the mountains or in the foothills. The potential for adaptation to drought of the Vercors and Villar d'Arène farms has been explored through semi-directed interviews, showing a diversity of possible responses depending on, among other factors, type of livestock production (sheep, suckler or dairy cattle), dependency on hay stocks, diversity of available grassland types, and potential to access new areas for grazing or hay making (Nettier et al. 2011).

Long-term ecological observations on plants and animals by ecologists and natural area managers make it possible to record changes in the fauna and flora and to relate these to past climate data monitored by climatologists. Experimental climate manipulations have also been set up in order to analyse the effects of extreme drought expected from climate scenarios on plant diversity, components of agronomic value and soil functioning. For instance, first results indicate a strong resistance of the studied subalpine grasslands on the Vercors Plateaux and at Lautaret to an extreme drought event, but the consequences of longer and/or even more intense events are unknown yet.

Scenarios of future farm system organisation and resulting changes in grassland management are developed together with agronomists, sociologists and stakeholders. Their impacts on social, economic and ecological components of the human-environment system will be assessed through participative approaches and modelling, (see Gaube and Haberl, Chap. 3 in this volume) in particular using ecosystem service models as presented above in the case of Villar d'Arène.

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20.4.3 Mutations of Alpine Tourism in the Face of Climate Change

Tourism is a major economic and social component in the territory of the Central French Alps LTSER Platform. Tourism activities have strong direct and indirect impacts on ecosystems, on which they also depend, especially in the case of winter sports. A new project therefore focuses on the coupled dynamics of tourism and ecosystems in the context of climate change. Alpine tourism is facing a severe structural crisis and its future is full of uncertainties. Climate change is one of them, and is often an accelerator of other crises (Elsasser and Bürki 2002). For many observers, the current tourism system established during the second half of the twentieth century, which is mainly concentrated on skiing, is largely exhausted and condemned to undergoing drastic changes. In this context, research questions regard two complementary levels and at the same time highlight current representations of alpine tourism and possible new actions for this sector: (i) What are the themes and indicators by which climate change is recognised by different stakeholders (visitors, actors from the tourism industry, members of the wildlife or administrative sectors)? (ii) What are the current observable impacts of climate change on recreation and professional tourism uses that are involved in outdoor activities? and (iii) To what extent and in what ways do actors from the tourism industry adapt their practices to ongoing and foreseeable changes due to rising concerns about climate change and biodiversity loss? Addressing such questions requires the strengthening of social sciences within the LTSER Platform and promotion of interdisciplinarity.

A survey questionnaire among individual participants of the LTSER Platform in order to identify ongoing and planned interdisciplinary projects on the interactions between ecosystems and tourism in the context of climate change highlighted the rarity of research on this question in spite of a broad recognition of its importance for the future of the study area. This survey also made it possible to build a better understanding within the Platform of ongoing projects on related topics and of their approaches and methods, and to identify research questions potentially conducive to collaboration between scientists from different disciplines. Based on this understanding, we called for internal proposals and two new projects are now due to start.

The first project focuses on the consideration of natural ecosystems around ski resorts, with a study of wetlands in Val Thorens (Savoie). Novel wetland management methods are being experimented with in order to preserve these ecosystems and, simultaneously, to use their potential to support tourism, both through their cultural value and as critically linked to snowmaking through competition for water supply. A survey will be carried out to analyse different approaches, their implementation and their ecological, economic and social impacts, as well as the role of different involved parties, including that of scientists specialising in wetland ecology. This research will involve one ecologist, one economist and one sociologist.

The second project analyses current and future linkages between tourism and natural ecosystems as seen by specialists of each of these sectors, using the Vercors Regional Natural Park as a case study. Semi-open interviews will be conducted

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with members of the tourism industry, ecologists and environmental advocates. This work will involve a geographer and a sociologist specialised in tourism and with a background in ecology.

20.5 Conclusion

Although the construction of a LTSER Platform requires much patience, only a short time after the creation of the Central French Alps LTSER Platform, individual participants and partner institutions conclude that it has provided an important framework to formalise already ongoing collaborations between different disciplines. Moreover, already within the first years of the Platform's existence, a process of de-compartmentalisation has started and had led to the initiation of several common projects in which social sciences are truly incorporated into the design of research agendas. In particular climate change has appeared as a common issue fostering transdisciplinary collaboration. Nevertheless, as in many similar long-term ecological Sites (Ohl et al. 2007), ecologists are still the most numerous group in our LTSER Platform. Incorporating a social sciences dimension into ecological research will ensure consistency of knowledge and the collection of data reflecting the complexity and changing nature of the coupled natural-human environment system, in order to enhance our ability to project global change impacts. The focus on ecosystem services and nature's values to land users is particularly promising in terms of enhancing integration. Research results will open new perspectives for management as well as for territorial development and policies.

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